

Solution Manual: Multivariate Delta Method Exercises

Sketch solutions emphasizing the key ideas and calculations.

1. Proof of the multivariate Delta Method

Use the first-order Taylor expansion $g(T_n) = g(\theta) + \nabla g(\theta)^T(T_n - \theta) + o_p(\|T_n - \theta\|)$. Multiply by $\sqrt{n}$ and apply Slutsky.

2. Ratio $g(x,y)=x/y$

Compute $\nabla g = (1/y, -x/y^2)$. Evaluate at $\theta = (\theta_1, \theta_2)$. The asymptotic variance is $\nabla g(\theta)^T \Sigma \nabla g(\theta)$.

3. Distribution of S_n/X_n

Apply the previous result with $g(\mu, \sigma) = \sigma/\mu$. Compute the gradient and substitute into the variance formula.

4. Coefficient of variation

The gradient is $(-\sigma/\mu^2, 1/\mu)$. Use $\nabla g^T \Sigma \nabla g$.

5. $\log(p_1/p_2)$

Take $g(x,y) = \log x - \log y$. Then $\nabla g = (1/x, -1/y)$. Evaluate at the population probabilities.

6. μ/σ^2

Use $g(x,y) = x/y$. The gradient is $(1/y, -x/y^2)$. Apply the Delta Method.

7. Correlation coefficient

Express the correlation as a smooth function of sample means, variances, and covariance. Use the multivariate theorem and compute the Jacobian.

8. $\exp(a^T \beta)$

The gradient is $\exp(a^T \beta) a$. The asymptotic variance is $\exp(2a^T \beta) a^T \Sigma a$.

9. $T_{1n}T_{2n}/T_{3n}$

Use $g(x,y,z) = xy/z$. Compute partial derivatives $(y/z, x/z, -xy/z^2)$.

10. Odds ratio

Take logarithms first. Apply the Delta Method to the log odds ratio, then exponentiate if desired.

11. Vector-valued theorem

Replace the gradient by the Jacobian matrix $J_g(\theta)$. The limit covariance becomes $J_g(\theta)\Sigma J_g(\theta)^T$.

12. Vanishing Jacobian

For $g(x,y)=x^2+y^2$ at $\theta=(0,0)$, $J_g(\theta)=0$. The first-order term disappears, so a second-order Taylor expansion is required.

Core Formulae

If $\sqrt{n}(\mathbf{T}_n - \theta) \Rightarrow N(0, \Sigma)$ and $g: \mathbb{R}^k \rightarrow \mathbb{R}$ is differentiable, then $\sqrt{n}(g(\mathbf{T}_n) - g(\theta)) \Rightarrow N(0, \nabla g(\theta)^T \Sigma \nabla g(\theta))$. For vector-valued g , replace the gradient by the Jacobian matrix.